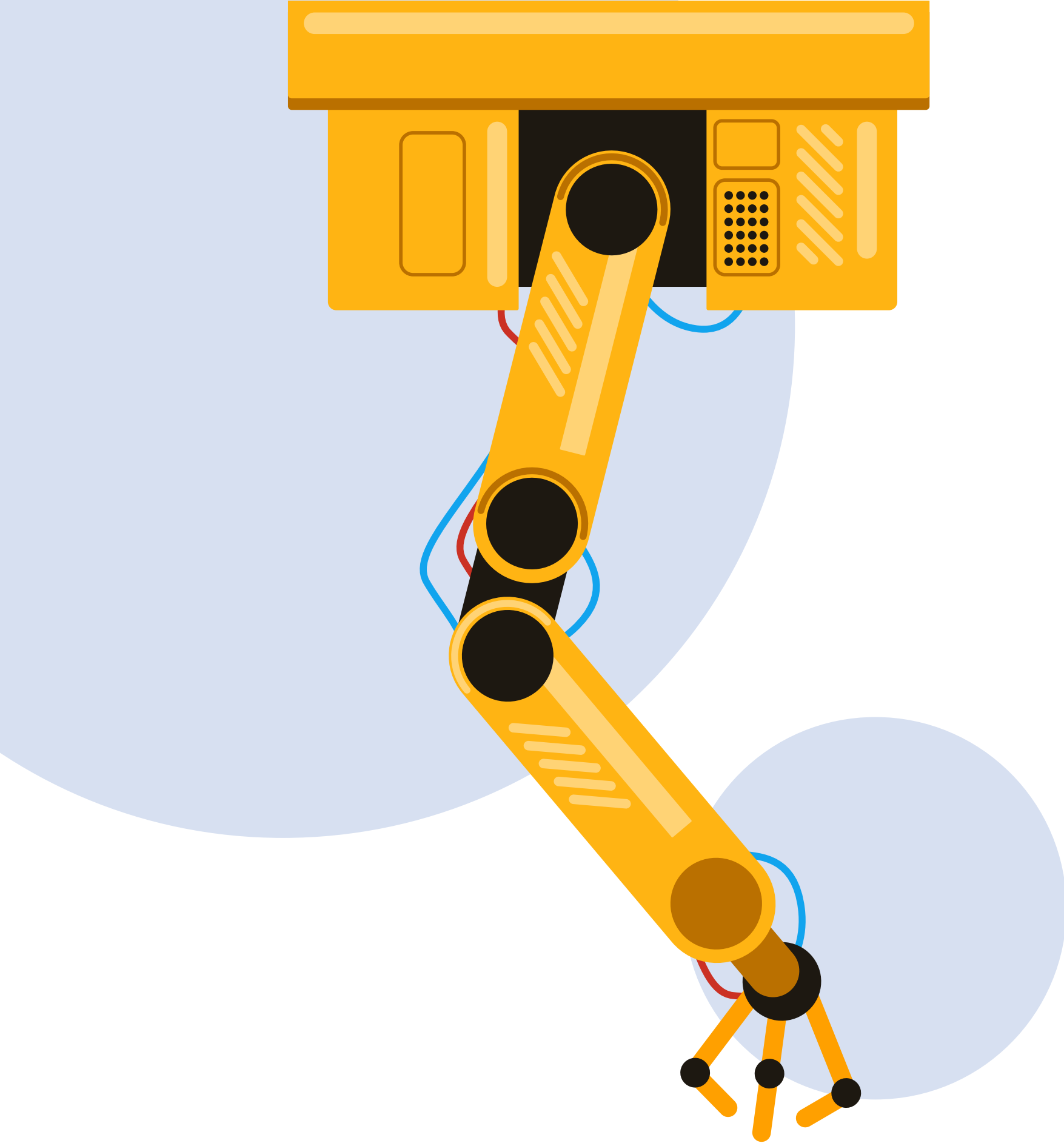


CLASE 10

Robótica

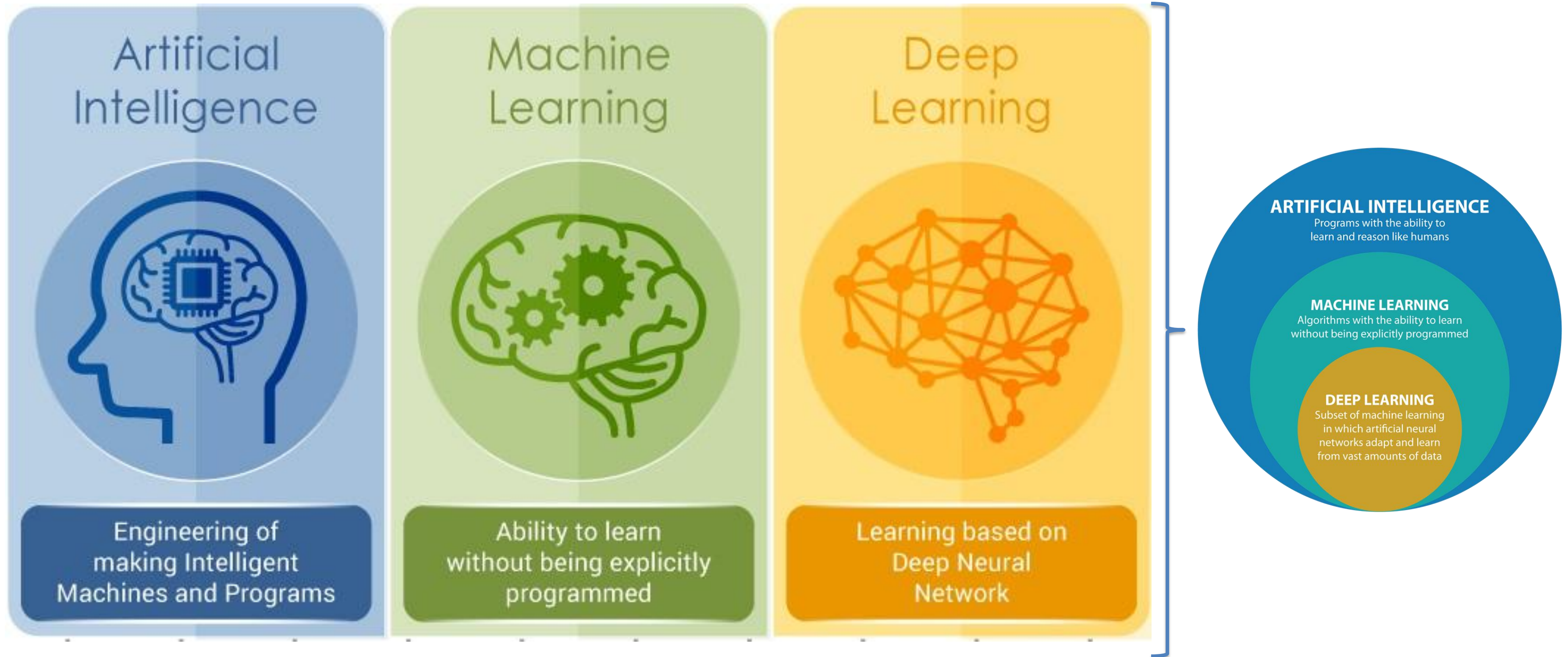




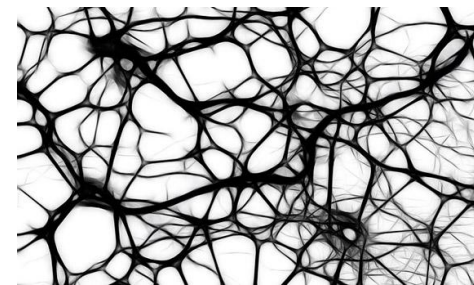
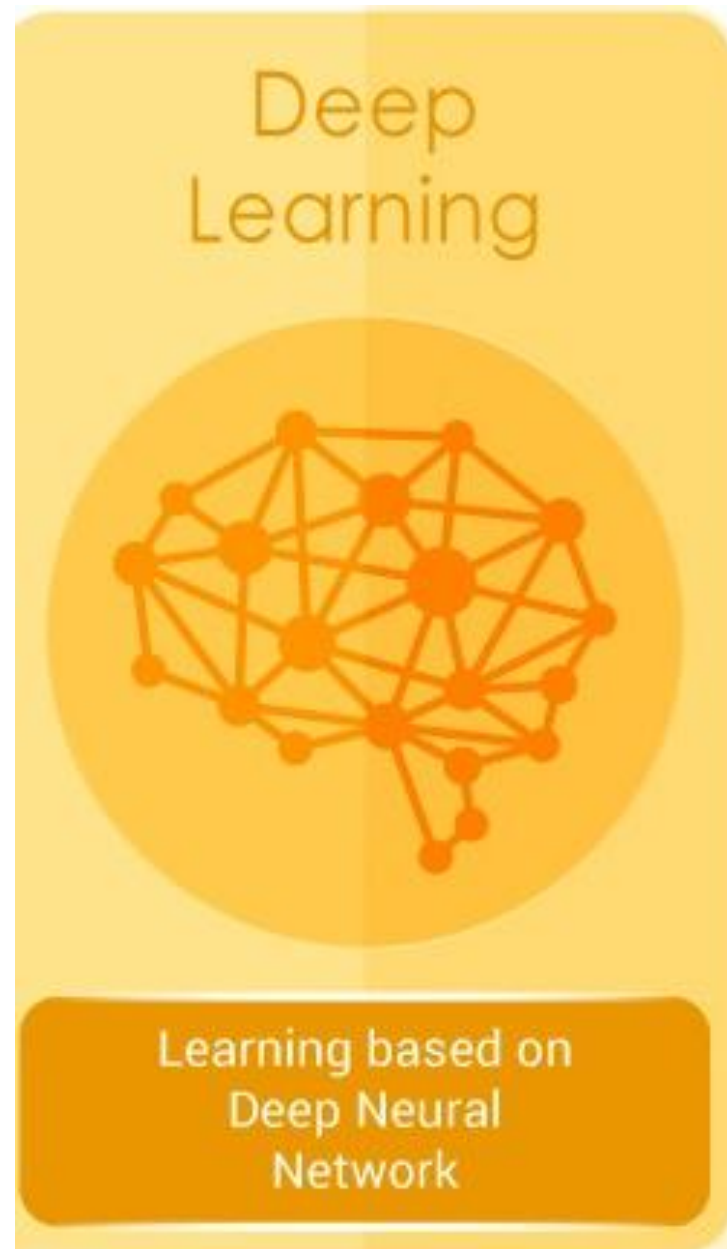
Inteligencia artificial (Teoría)

- A.1 Historia e introducción
- A.2 Clasificación de redes
- A.3 Red neuronal convolucional
- A.4 Métodos de evaluación

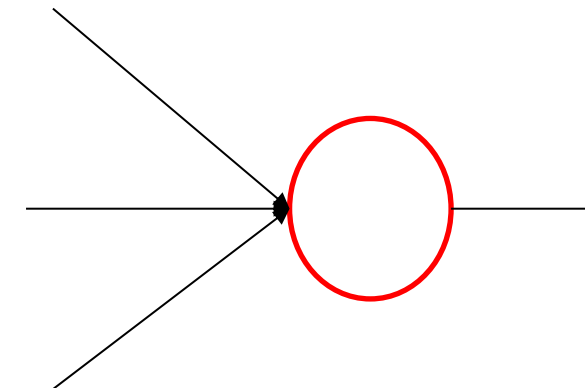
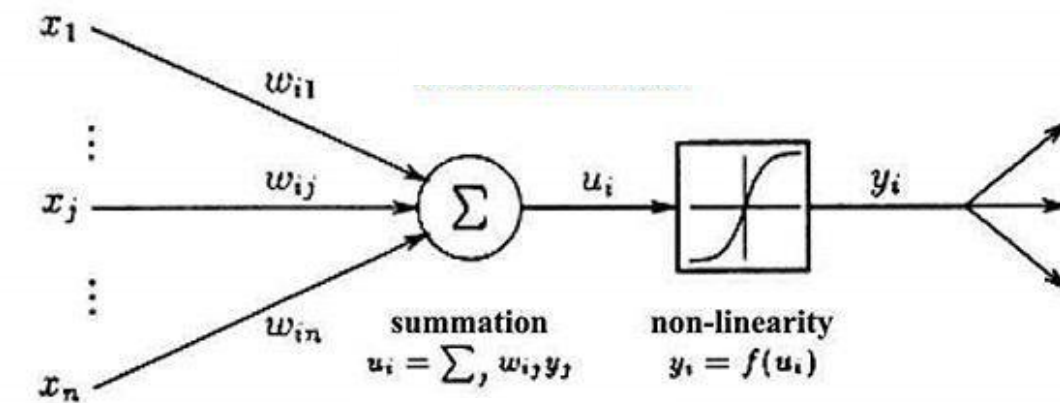
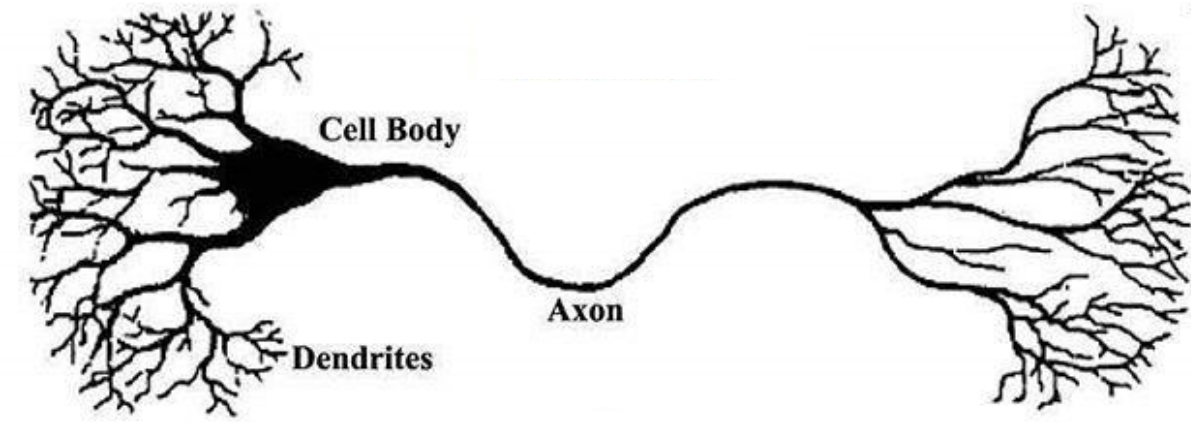
Historia e introducción



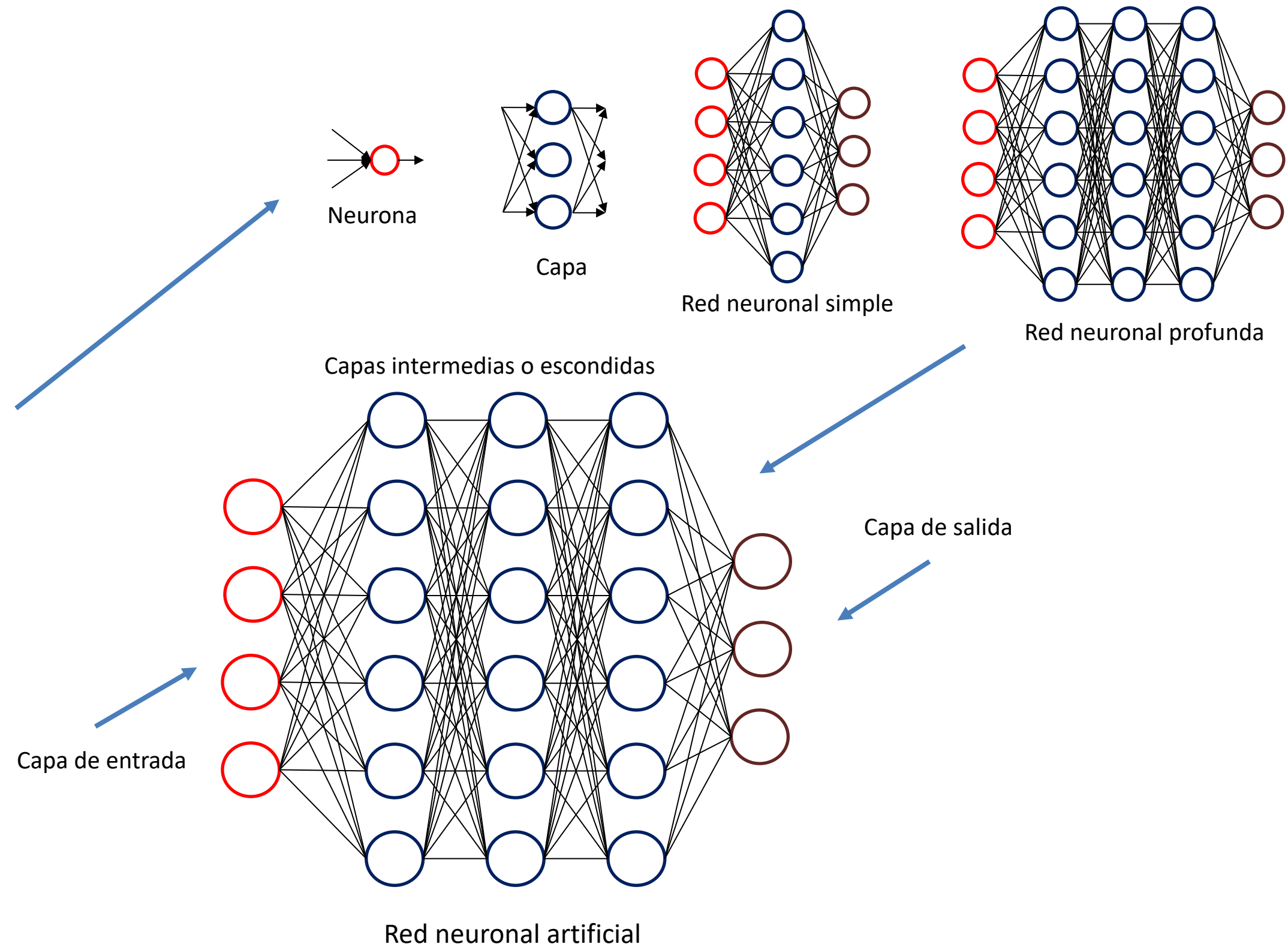
Historia e introducción



Neuronas Biológicas



Historia e introducción



Clasificación

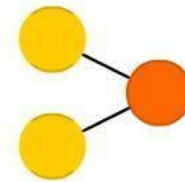
A mostly complete chart of

Neural Networks

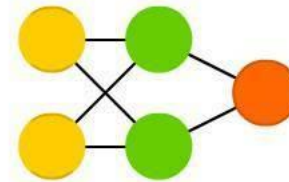
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- Backfed Input Cell
- Input Cell
- Noisy Input Cell
- Hidden Cell
- Probablistic Hidden Cell
- Spiking Hidden Cell
- Output Cell
- Match Input Output Cell
- Recurrent Cell
- Memory Cell
- Different Memory Cell
- Kernel
- Convolution or Pool

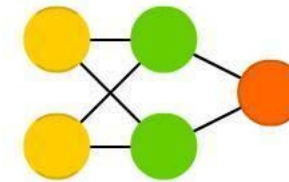
Perceptron (P)



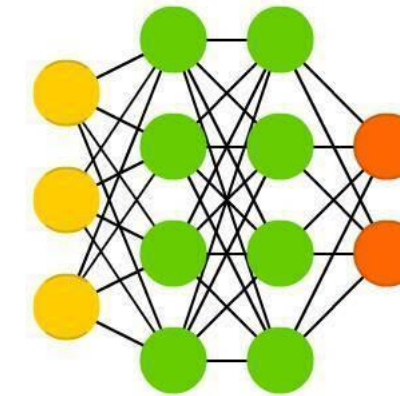
Feed Forward (FF)



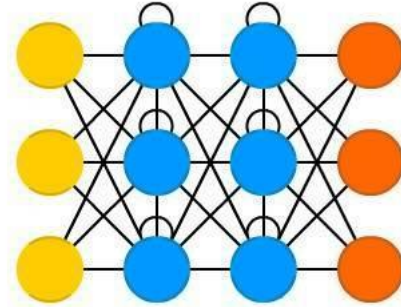
Radial Basis Network (RBF)



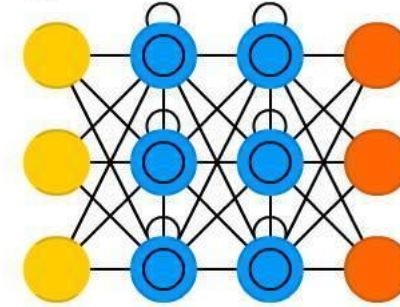
Deep Feed Forward (DFF)



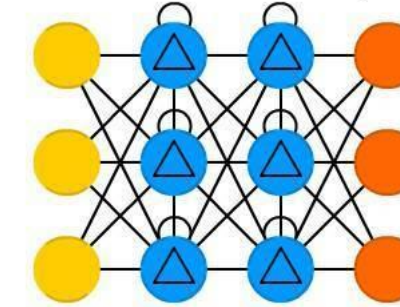
Recurrent Neural Network (RNN)



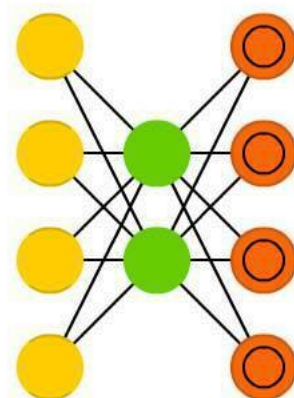
Long / Short Term Memory (LSTM)



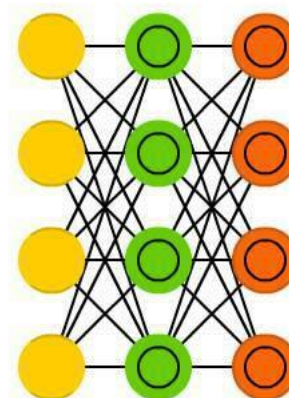
Gated Recurrent Unit (GRU)



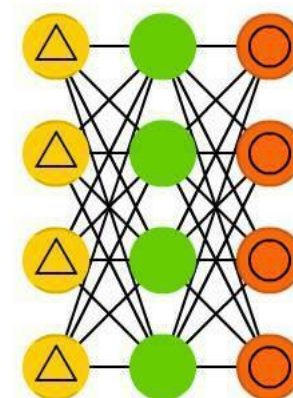
Auto Encoder (AE)



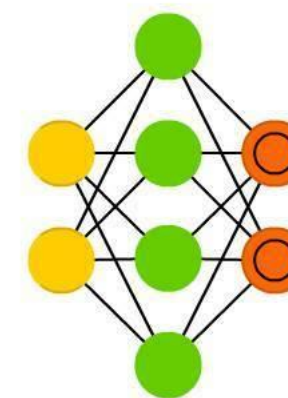
Variational AE (VAE)



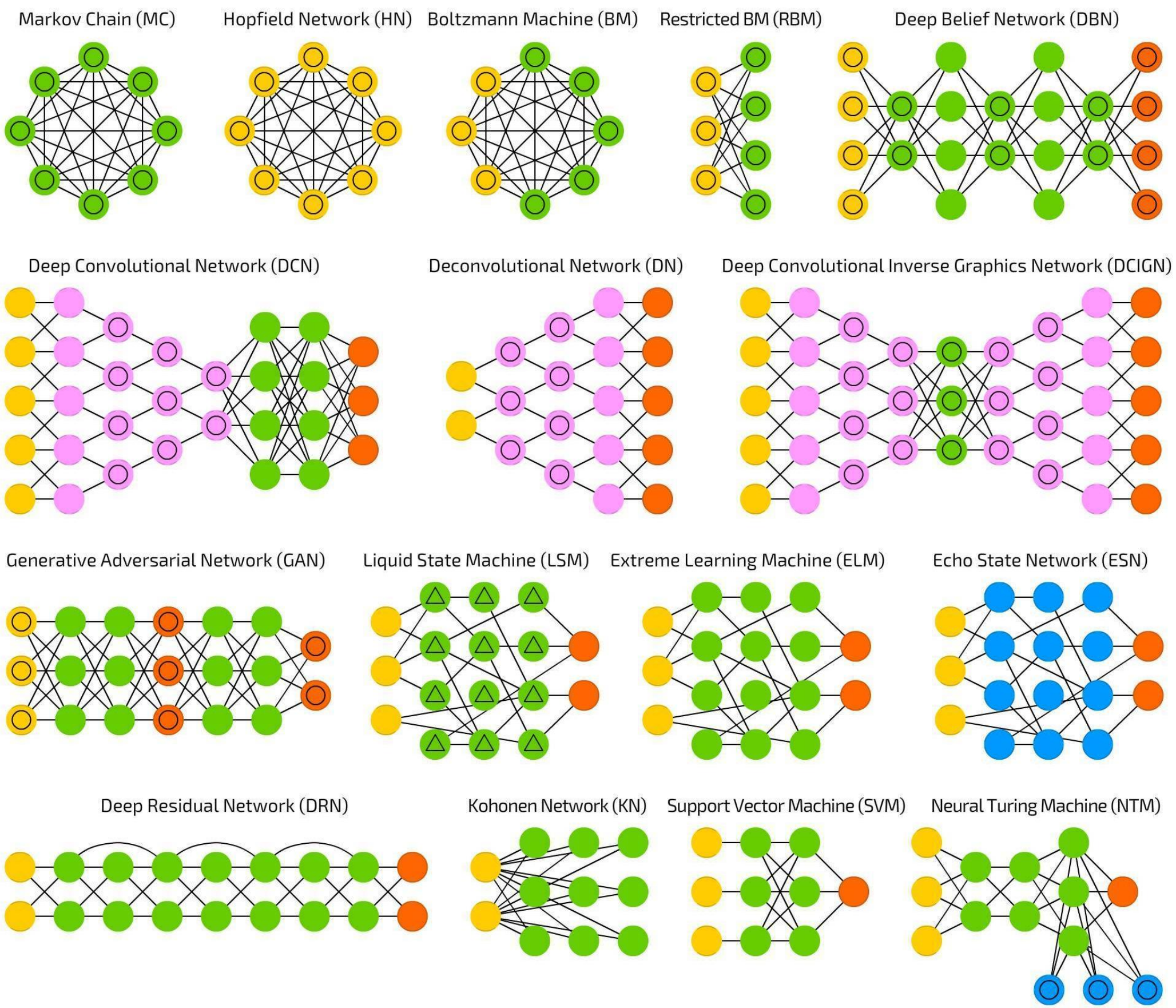
Denoising AE (DAE)



Sparse AE (SAE)



Clasificación



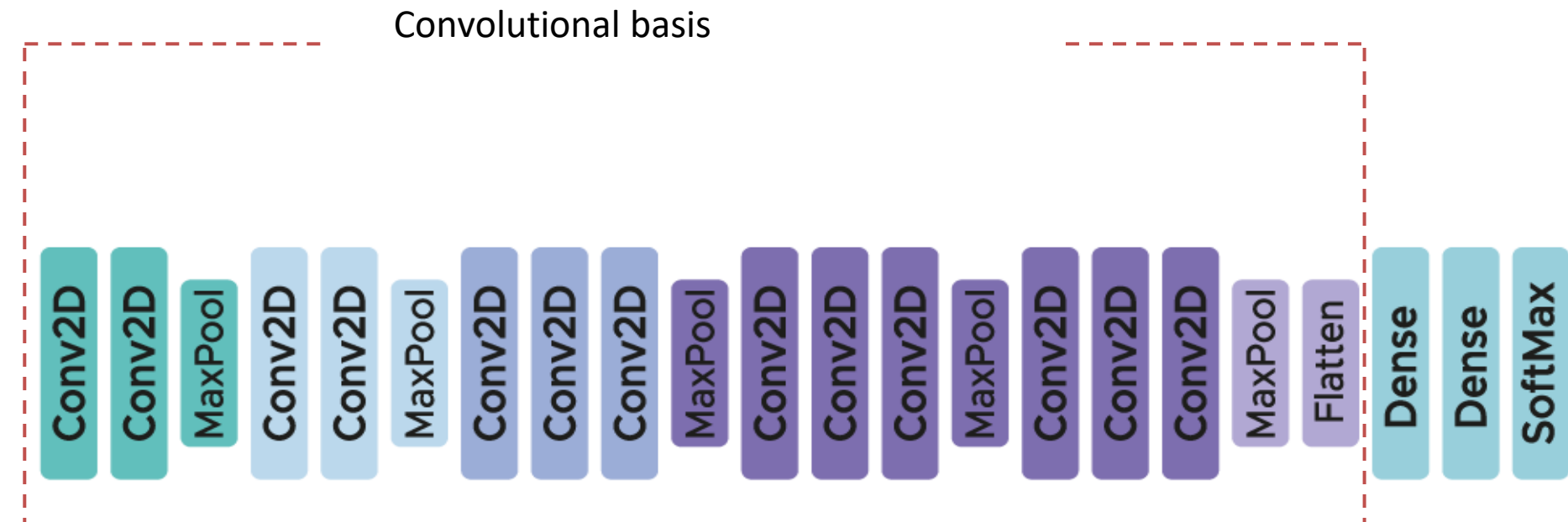
Red neuronal convolucional CNN

Convolutional blocks of a pre-trained CNN

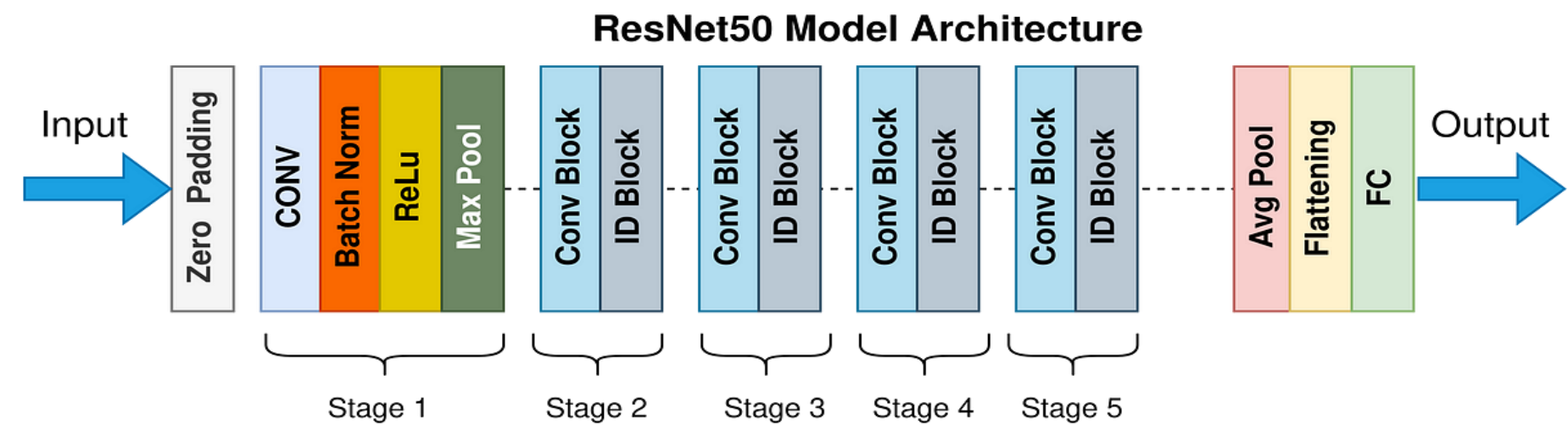
- VGG16
- ResNet50
- MobileNet
- InceptionResNetV2
- GoogleNet
- AlexNet



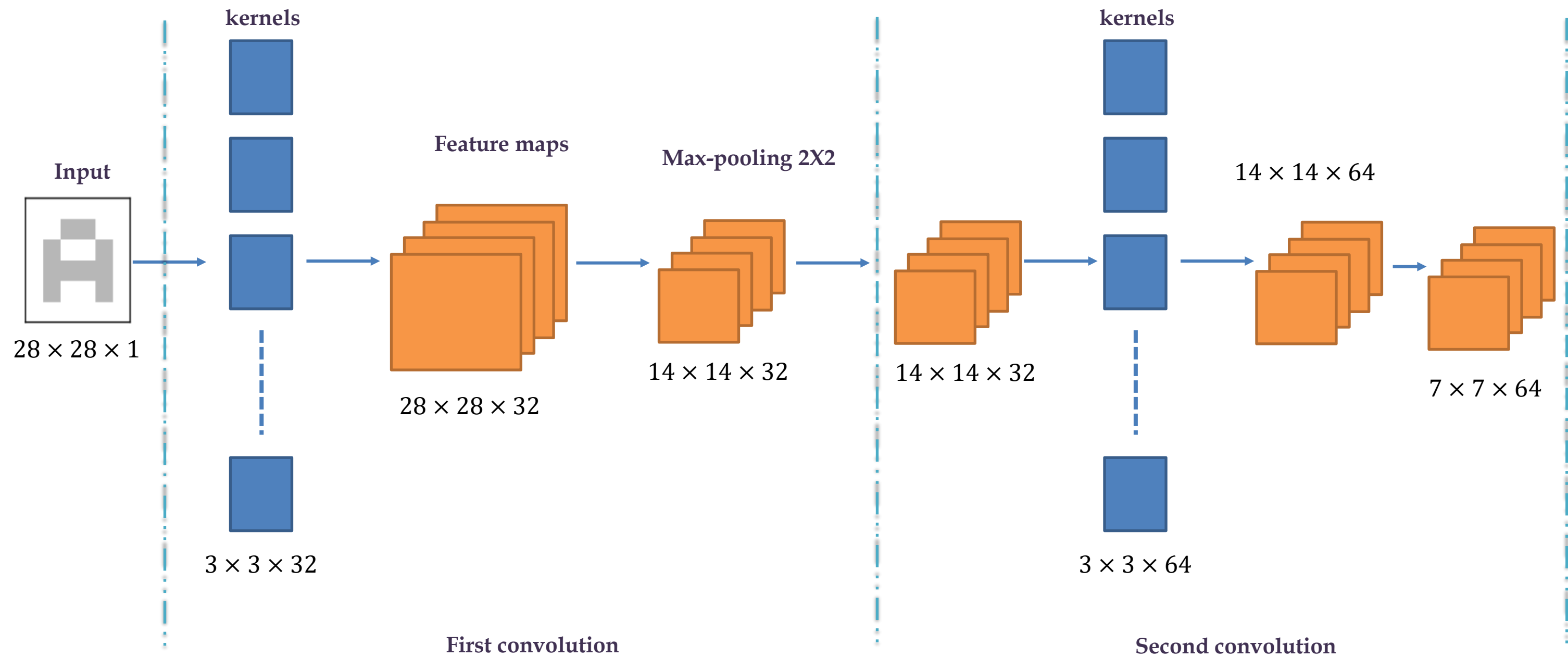

Transfer learning



VGG16 Architecture

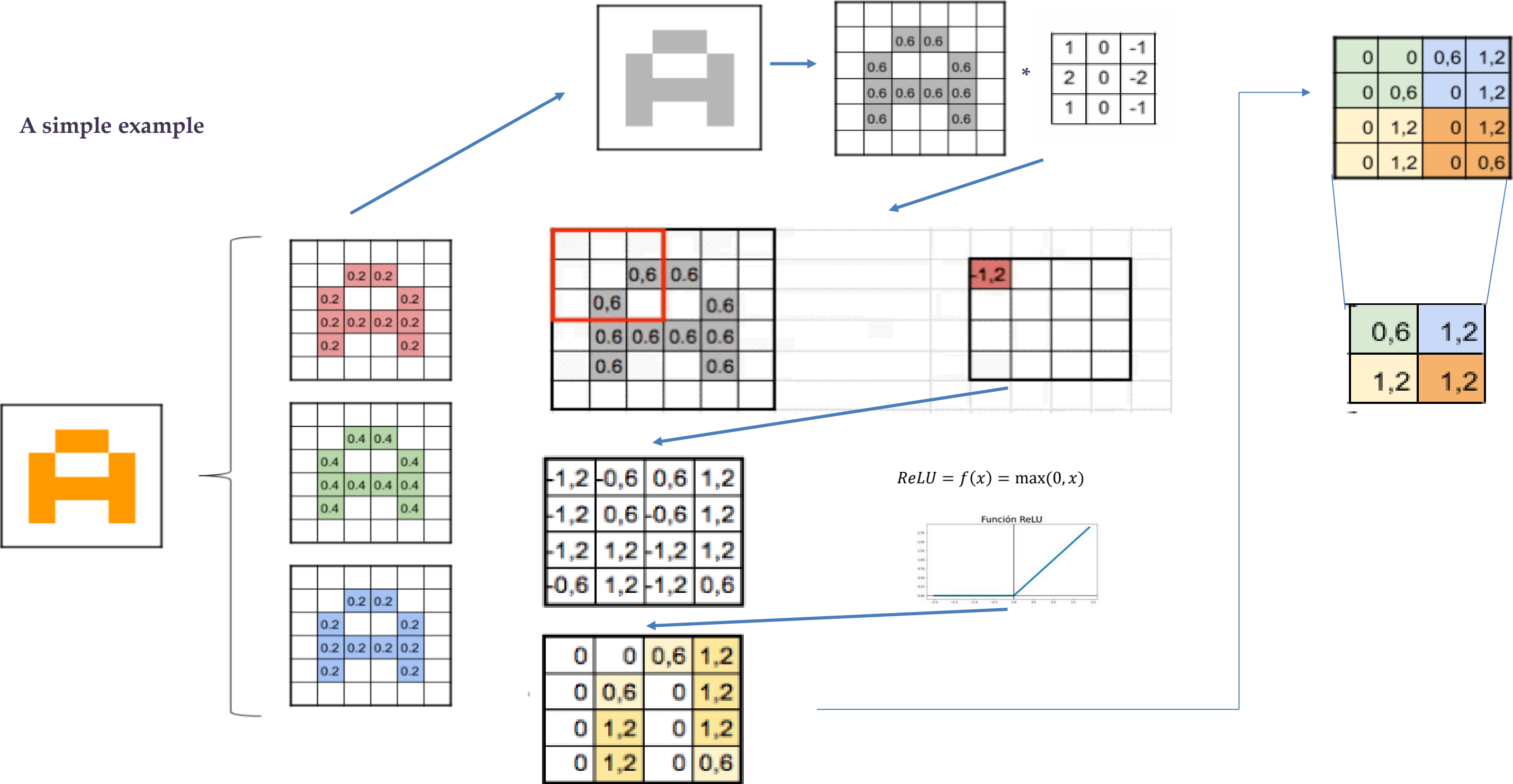


Red neuronal convolucional CNN



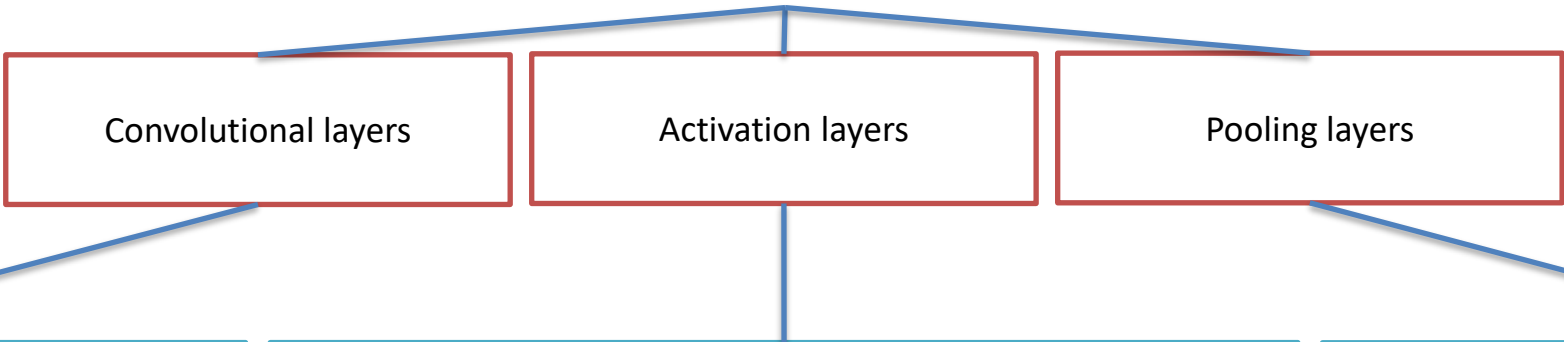
Red neuronal convolucional CNN

A simple example



Red neuronal convolucional CNN

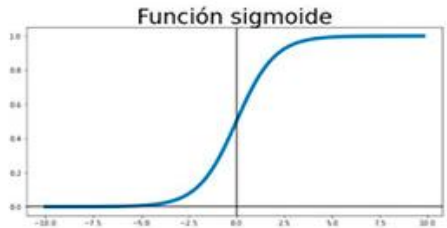
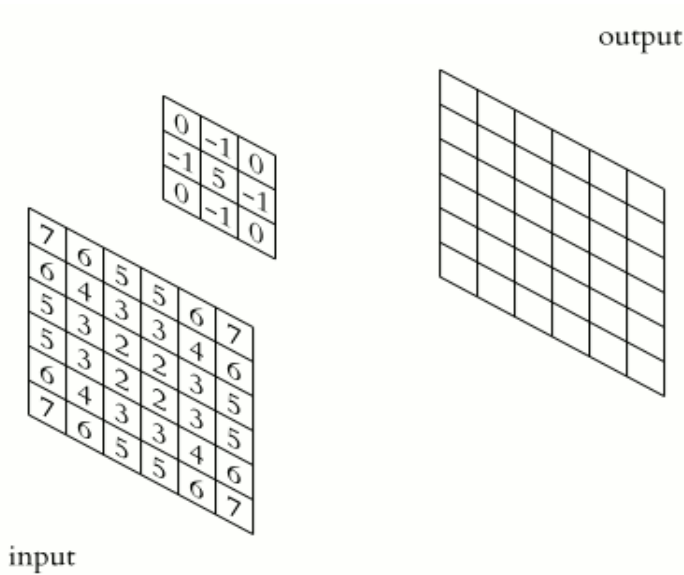
Convolutional blocks



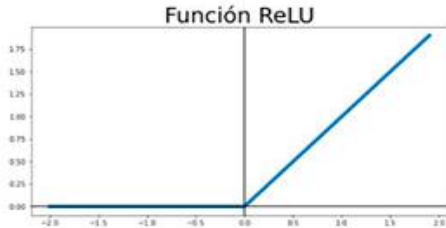
These layers are responsible for extracting important features from the image. They use special filters called "kernels" that slide across the image and perform mathematical operations to highlight patterns such as edges, textures, and shapes.

After convolution, an activation function is applied which introduces nonlinearity into the network. The activation function determines whether a neuron is activated or not, which helps capture complex relationships in the data.

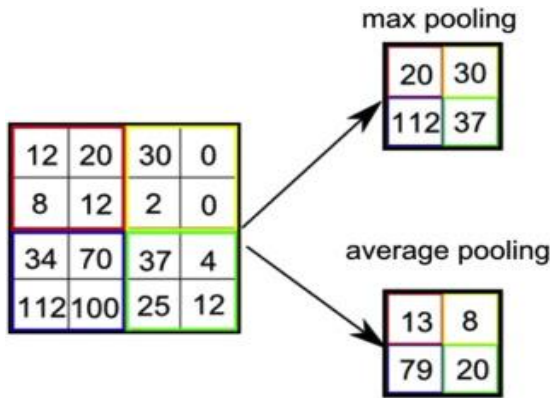
These layers reduce the dimensionality of the features obtained by the convolutional layers. The pooling process combines the information into small regions to summarize it and retain only the most relevant information.



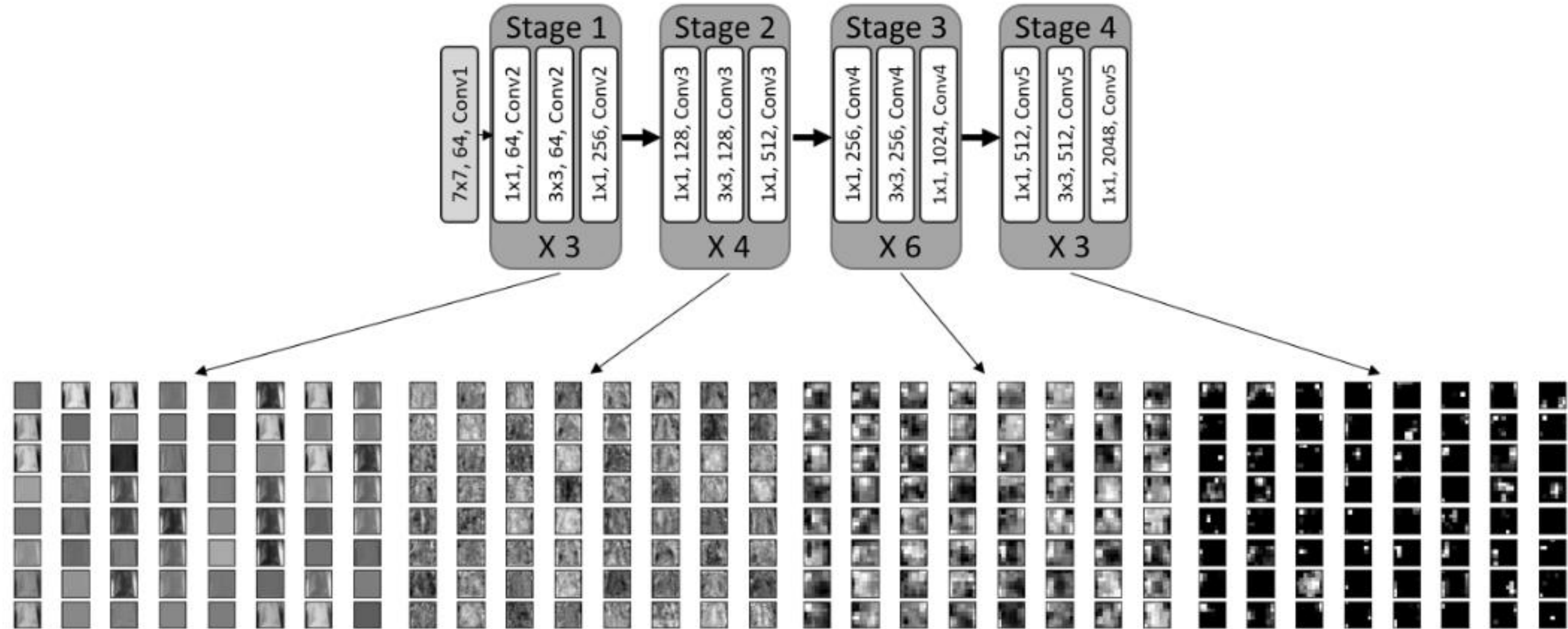
$$\sigma(x) = \frac{1}{1 + e^{-x}}$$



$$f(x) = \max(0, x)$$



Red neuronal convolucional CNN



Métodos de evaluación

Clasificadores

- Support Vector Machine (SVM)
- Logistic regression (LR)
- Nearest neighbors (KNN)
- Naïve-Bayes (NB)
- Nearest centroid (NC)

Matriz de confusión

		Predicted	
		Negative	Positive
Actual	Negative	True Negative (TN)	False Positive (FP)
	Positive	False Negative (FN)	True Positive (TP)

$$\begin{aligned} \text{Precision} \\ &= \frac{TP}{TP + FP} \end{aligned}$$

$$\begin{aligned} \text{Recall(sensitivity)} \\ &= \frac{TP}{TP + FN} \end{aligned}$$

$$\begin{aligned} \text{F1 - Score} \\ &= 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \end{aligned}$$

$$\begin{aligned} \text{Accuracy} \\ &= \frac{TP + TN}{TP + TN + FP + FN} \end{aligned}$$

$$\begin{aligned} \text{Specificity} \\ &= \frac{TN}{TN + FP} \end{aligned}$$